

Lead Enrichment Strategy & Process

End-to-End Guide — LaunchHouse Events / Sales Outreach Platform

Last updated: April 25, 2026

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1. Executive Summary

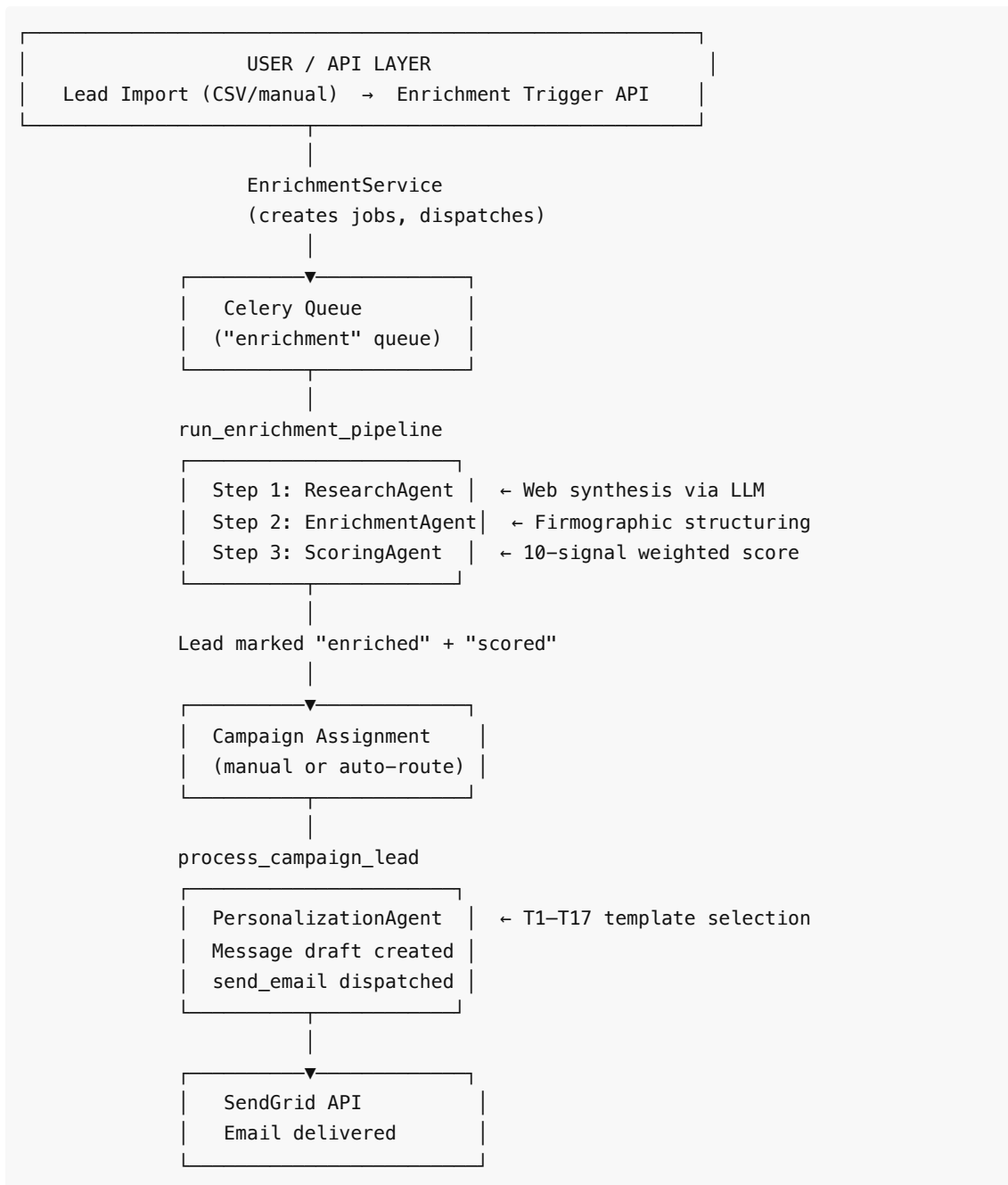
The Sales Outreach Platform automates end-to-end outbound prospecting for **LaunchHouse Events**, a Cvent development and professional services firm. The platform ingests raw lead lists (CSV/manual), runs a fully automated 3-stage AI enrichment pipeline, scores each lead for fit, and generates hyper-personalized email sequences using templates calibrated to the events industry.

Core value proposition: Turn a cold CSV of company names and emails into a ranked, personally researched, and actively-sequenced outbound campaign with zero human effort between import and first email draft.

Technology foundations:

- Primary AI: Anthropic Claude (`claude-sonnet-4-6`)
 - Orchestration: Celery (distributed task queue) + Redis broker
 - Storage: PostgreSQL with multi-tenant row-level isolation
 - Email delivery: SendGrid
 - Backend: Python / FastAPI (async)
 - Frontend: Next.js 14
-

2. System Architecture Overview



3. Data Model

Core Tables

Table	Purpose	Key Fields
companies	Firmographic data per company	name, domain, industry, employee_count, revenue_range, location, description, technologies (JSONB)
contacts	Individual people	email, first_name, last_name, title, department, phone, linkedin_url, location, timezone

leads	The join between contact + company with pipeline state	status, enrichment_status, source, tags, enriched_at
lead_scores	AI-generated score per lead	overall_score (float), tier (hot/warm/cold), signal_scores (JSONB), explanation, model_used
enrichment_jobs	Audit trail per enrichment step	job_type, status, provider, input_data, output_data, error, duration_ms, tokens_used, cost_usd
research_data	Raw research output	source, url, title, content, relevance_score, metadata (JSONB)
enrichment_data	Structured enrichment output	data_type, provider, data (JSONB), confidence, version
ai_insights	Human-readable AI summaries	insight_type, content, confidence, model_used, tokens_used
lead_activities	Audit log of all events	activity_type, title, description, metadata (JSONB)
import_batches	CSV upload tracking	file_name, status, total_rows, processed_rows, success_rows, error_rows

Campaign Tables

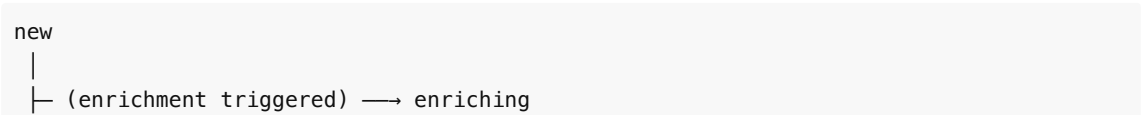
Table	Purpose
campaigns	Campaign definition with sequence steps, sender, status
campaign_leads	Lead × Campaign assignment with current_step, status
messages	Every email draft/sent — subject, body_html, body_text, status, ai_generated, personalization_hooks
follow_ups	Scheduled next-step email queue
replies	Inbound reply tracking with intent classification
email_events	Open/click/bounce/sent tracking from SendGrid webhooks

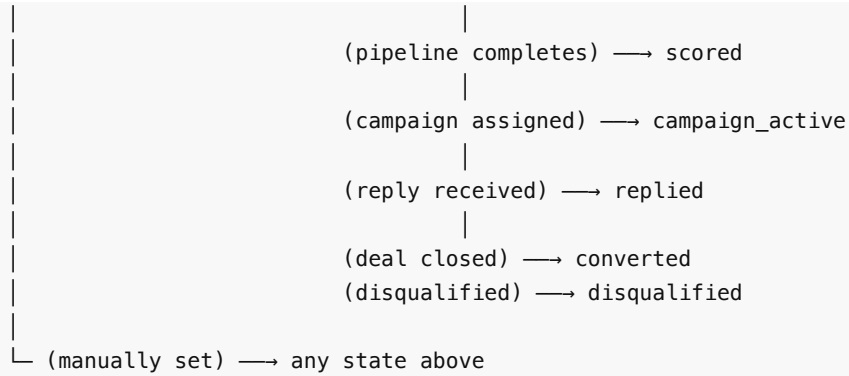
All Entities Are Multi-Tenant

Every table includes a `tenant_id` column linked to the `tenants` table. Row-level isolation is enforced at the middleware and query layer — tenants cannot see each other's data.

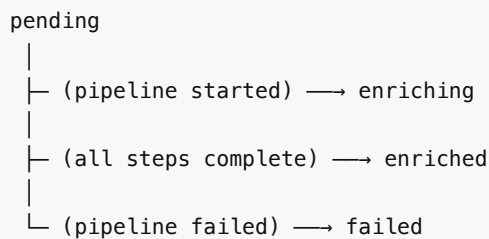
4. Lead Lifecycle & Status Transitions

lead.status Flow

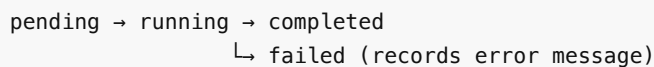




Lead.enrichment_status Flow



enrichment_job.status per step



5. Phase 1 — Lead Import & Registration

How Leads Enter the System

Leads are imported in two ways:

1. **CSV Upload** — via the `/api/leads/import` endpoint. Columns are mapped to `Company` and `Contact` fields. Each row creates a `Company` + `Contact` + `Lead` record. Batch tracked in `import_batches`.
2. **Manual Entry** — via the UI form or REST API. Single point-of-entry creating the same three records.

Initial Data Required

- Company name (required)
- Contact email (required)
- First/last name (optional but strongly recommended)
- Job title (optional)
- Company domain (optional — improves research quality significantly)

Post-Import State

- `lead.status = "new"`
- `lead.enrichment_status = "pending"`
- Company, Contact records created with whatever raw data was available
- No enrichment, scoring, or AI processing has occurred yet

Triggering Enrichment

Enrichment is triggered explicitly — either:

- Manually: user selects leads in the UI and clicks "Enrich"
- Via API: `POST /api/enrichment/enrich` with `lead_ids[]` and `enrichment_types[]`

Default enrichment types requested: `["web_research", "company", "scoring"]`

6. Phase 2 — Web Research Agent

Class: `ResearchAgent` (`agents/research_agent.py`)

LLM Temperature: 0.3 (slight creativity for synthesis)

Model: Claude `claude-sonnet-4-6` via `BaseAgent.get_llm()`

Purpose

Gather publicly available intelligence about the company and contact: news, events, technology stack, funding, and buying signals — all without any human effort.

Search Strategy — 5 Queries Built per Lead

Query #	Pattern	Goal
1	<code>{company_name} company news recent</code>	Latest company updates, announcements, changes
2	<code>{company_name} events Cvent</code>	Cvent usage evidence, event calendar, hosted events
3	<code>{company_name} technology stack</code>	Technology platforms in use
4	<code>site:{domain}</code>	Homepage / about page / events page scan
5	<code>{contact_name} {company_name} LinkedIn</code>	Contact seniority, career path, responsibilities

Data Sources (Current Status)

Source	Status	Purpose
Tavily	✅ Live	Primary web search — AI-optimised snippets, used for all 5 per-lead queries
Firecrawl (search)	✅ Live (fallback)	Full-page search fallback if Tavily returns no results
Firecrawl (scrape)	✅ Live	Homepage scrape — always runs if domain is known, returns full markdown (capped 3 000 chars)
Claude LLM	✅ Live	Synthesis and structuring of all search findings into <code>ResearchOutput</code>

Provider priority: Tavily is tried first for every query. If it returns results, Firecrawl search is skipped. Firecrawl scrape runs independently in parallel for the homepage regardless of search results.

Python packages required: `tavily-python>=0.3.0` , `firecrawl-py>=0.0.16` (both in `requirements.txt` and installed in all Docker containers).

Research Output Schema (`ResearchOutput`)

```
{
  "company_summary": "2-3 sentence overview of the company",
  "recent_news": [
    { "title": "...", "url": "...", "date": "...", "summary": "..." }
  ],
  "key_people": [
    { "name": "...", "title": "...", "linkedin_url": "..." }
  ],
  "technology_stack": ["Cvent", "Salesforce", "Marketo"],
  "funding_info": {
    "stage": "Series B",
    "amount": "$50M",
    "investors": ["Sequoia", "..."]
  },
  "industry_signals": [
    "company posted 3 event manager roles in past 30 days"
  ],
  "events_attended": [
    {
      "event": "Dreamforce 2024",
      "year": 2024,
      "month": 9,
      "date_label": "September 2024",
      "type": "past",
      "role": "sponsor",
      "confirmed": true,
      "url": "https://...",
      "description": "Gold sponsor, hosted networking dinner"
    }
  ],
  "competitor_info": ["Competitor A uses similar platform"],
  "relevance_score": 0.82
}
```

Events Attended — Classification

The `events_attended` array is the most critical research output for personalization. Each event is classified across two dimensions:

`type` values:

- `past` — event has already occurred
- `upcoming` — event is in the future

- `recurring` — annual/regular event

role values:

- `attendee` — company sent staff to the event
- `sponsor` — company financially sponsored the event
- `host` — company ran/organized the event
- `speaker` — company had a speaker at the event
- `unknown` — role unclear from available data

Storage

Research output is stored in two places:

1. `research_data` table — raw `source="ai_synthesis"` record with full JSON content
2. `ai_insights` table — `insight_type="research_summary"` with human-readable summary text

The corresponding `enrichment_jobs` record (type= `web_research`) is marked `completed` on success, `failed` with error text if an exception occurs.

7. Phase 3 — Company & Contact Enrichment Agent

Class: `EnrichmentAgent` (`agents/enrichment_agent.py`)

LLM Temperature: 0.2 (precision-focused, minimal hallucination)

Model: Claude `claude-sonnet-4-6`

Purpose

Take the raw lead data + Phase 2 research output and produce structured firmographic and contact-demographic data. Determine the company's exact profile and assess the contact's decision-making authority.

Inputs

1. Raw lead data (company name, domain, contact name/title, any existing DB fields)
2. Full `ResearchOutput` from Phase 2

Enrichment Output Schema

Company block:

```
{
  "employee_count_range": "51-200",
  "revenue_range": "$10M-$50M",
  "industry": "Technology / SaaS",
  "sub_industry": "Event Management Software",
  "founded_year": 2018,
  "headquarters": "San Francisco, CA",
  "description": "Cloud-based event operations platform for enterprise marketing teams",
  "technologies": ["Cvent", "Salesforce", "HubSpot", "Zoom"]
}
```

Contact block:

```
{
  "seniority": "Director",
  "department": "Events",
  "likely_responsibilities": [
    "Manages Cvent license and configuration",
    "Coordinates with marketing on event calendar",
    "Owns vendor selection for event tech"
  ],
  "decision_maker": true,
  "buyer_persona": "Events Operations Lead"
}
```

Confidence score: 0.0–1.0 (how confident the agent is in its output given available data)

Database Write-Back

After enrichment, the pipeline **updates existing company and contact records** with the enriched data:

- `company.industry` ← `enrichment.company.industry`
- `company.description` ← `enrichment.company.description`
- `contact.department` ← `enrichment.contact.department`
- `contact.title` ← `enrichment.contact.seniority` (if not already set)

These write-backs ensure the main records stay current and that the UI displays enriched values everywhere.

Storage

1. `enrichment_data` table — `data_type="company_contact"` , `provider="anthropic"` , full JSON in `data` column
2. `ai_insights` table — `insight_type="company_enrichment"` with narrative summary

8. Phase 4 — AI Lead Scoring Engine

Class: `ScoringAgent` (`agents/scoring_agent.py`)
LLM Temperature: 0.2 (deterministic scoring)
Model: Claude `claude-sonnet-4-6`

Purpose

Assign a numeric fit score (0–100) and tier (Hot/Warm/Cold) to each lead, enabling sales prioritization and automated campaign routing.

Inputs

1. Lead data (contact + company fields)
2. `EnrichmentOutput` from Phase 3
3. `ResearchOutput` from Phase 2

The 10 Scoring Signals

#	Signal Name	Weight	What It Measures
---	-------------	--------	------------------

1	Company Size Fit	15%	Does the company's headcount match LaunchHouse's sweet spot (event teams that need outside help but aren't enterprise enough for full agencies)?
2	Industry Fit	12%	Is the company in a Cvent-heavy vertical? (Corporate events, association management, pharma, finance, tech conference orgs)
3	Title / Seniority Fit	15%	Is the contact a decision-maker or influencer for Cvent builds? (Director/VP/Head of Events, Marketing Operations, IT)
4	Technology Fit	10%	Does the company use Cvent or adjacent event tech (Aventri, RainFocus, bizzabo)? Any mention of event tech modernization?
5	Recent Activity Signals	12%	Hiring for event roles, conference announcements, new office/expansion, increased event calendars
6	Funding / Growth	8%	Recent funding rounds often precede expanded event calendars (investor days, road shows, conferences)
7	Event Usage (Cvent)	10%	Direct evidence of Cvent usage — sponsor/host/speaker at Cvent-powered events, Cvent case studies, Cvent community mentions
8	Engagement History	8%	Prior interaction with LaunchHouse (email opens, clicks, past replies, known relationship)
9	Timing Signals	5%	Budget cycle indicators (Q4 planning, fiscal year end), contract renewal timing, recent Cvent RFP activity
10	Geographic Fit	5%	US-based preferred (primary delivery market). North America / UK secondary.

Total weights sum to 100%.

Score Calculation

For each signal, Claude assigns a sub-score from **0.0 to 1.0** with reasoning. The final `overall_score` is:

$$\text{overall_score} = \sum_{i=1}^{10} \text{signal_score}_i \times \text{weight}_i \times 100$$

Example:

- Company Size: $0.8 \times 0.15 = 0.12$
- Industry Fit: $0.9 \times 0.12 = 0.108$
- Title/Seniority: $0.7 \times 0.15 = 0.105$
- Technology Fit: $1.0 \times 0.10 = 0.10$
- (... sum all 10 ...)
- **Final score: 72/100 → Warm tier**

Tier Definitions

Tier	Score Range	Sales Action	Color
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 Hot	75–100	Prioritize immediately — highest probability ICP match	Red/Crimson
 Warm	50–74	Moderate interest — nurture with personalized outreach	Amber
 Cold	0–49	Low probability — consider for future nurturing or disqualify	Blue/Gray

Scoring Output Schema (`ScoringOutput`)

```
{
  "overall_score": 72.4,
  "tier": "warm",
  "signals": [
    {
      "signal_name": "Technology Fit",
      "score": 0.90,
      "weight": 0.10,
      "reasoning": "Company mentioned Cvent in two job postings and LinkedIn bio references 'Cvent Community'"
    }
  ],
  "explanation": "Strong technology fit with confirmed Cvent usage. Director-level contact in Events team with likely purchasing authority. Company is 200–500 employees in growth phase with recent Series B – typical pre-expansion event calendar growth. Scored warm due to no direct recent engagement and uncertain timing.",
  "recommended_action": "Add to warm nurture sequence. Lead with T8 template if upcoming events confirmed, otherwise T1/T6."
}
```

Storage

1. `lead_scores` table — `overall_score`, `tier`, `signal_scores` (full JSON), `explanation`, `model_used="claude"`
2. `ai_insights` table — `insight_type="lead_score"` with narrative
3. `lead.status` updated to `"scored"`

9. Phase 5 — Personalization & Email Generation

Class: `PersonalizationAgent` (`agents/personalization_agent.py`)

Model: Claude `claude-sonnet-4-6`

Template Library: 17 templates (T1–T17)

Purpose

Generate hyper-personalized, on-brand cold email copy by selecting the right template from the playbook and filling it with lead-specific context. The output is a review-ready email draft — not a robo-blast.

Context Provided to the Agent

The agent receives:

1. **Lead data** — contact name, title, company, industry, location
2. **Research data** — events attended, recent news, technology stack, buying signals (from Phases 2–3)
3. **Sender info** — first name, calendar link, company name (LaunchHouse Events)
4. **Step config** — which sequence step this is (1–4)
5. **Previous email subject** — for reply-handler continuity
6. **Reply intent** — if the lead has replied, the classified intent (positive/not-now/wrong-contact/etc.)

Template Selection Hierarchy (Strict Priority Order)

The agent follows this decision tree — the **first matching condition wins**:

1. Reply received?

└ Intent = positive interest → T10 (Positive Interest Reply)

└ Intent = "send more info" → T11 (Send More Info Reply)

└ Intent = "not now" / deferral → T12 (Not Now Reply)

└ Intent = "already have support" → T13 (Already Have Support Reply)

└ Intent = wrong contact / referral → T14 (Wrong Contact / Referral Ask)

2. No reply – upcoming event confirmed?

└ Event 0–30 days away → T7 (Rush Framing)

└ Event 31–120 days away → T8 (Build Scoping – highest converting)

└ Event 120+ days away / past event → T5 (Event Trigger)

3. Recent news ≤45 days + events inferable → T9 (News Trigger)

4. No event/news but activity inferable → T6 (Fit-Based)

5. Core sequence steps (default):

└ Step 1 → T1 (Initial Outreach)

└ Step 2 → T2 (Value Add Follow-Up)

└ Step 3 → T3 (Bump – "worth a reply?")

└ Step 4 → T4 (Break-Up / Final Touch)

6. Scenario templates (special cases):

└ Voicemail left → T15 (Voicemail Follow-Up)

└ Meeting just booked → T16 (Meeting Confirmation)

└ Post-call → T17 (Post-Call Recap)

The 17 Templates — Summary

Core Sequence

Template	Name	Timing	CTA
T1	Initial Outreach	Day 1	One-pager or 15-min call
T2	Value Add Follow-Up	Day 4	Share useful content + light ask
T3	Bump	Day 9	"Worth a reply or not?"
T4	Break-Up	Day 20	Final touch → moves to nurture

Event & News Triggers

Template	Name	Condition	Key Framing
T5	Event Trigger	Verified event, any date	Event-specific Cvent build angle
T6	Fit-Based	No event found, activity inferable	Industry/motion overflow capacity
T7	Upcoming Event — Rush	0–30 days to event	Rush capacity, same-day turnaround
T8	Upcoming Event — Build	31–120 days	Fixed-fee build scoping (highest-converting)
T9	News Trigger	News ≤45 days + event implication	Congratulate + anticipate event expansion

Reply Handlers

Template	Name	Trigger
T10	Positive Interest	Lead expressed interest
T11	Send More Info	Lead asked for details
T12	Not Now	Lead deferred
T13	Already Have Support	Lead has existing vendor
T14	Wrong Contact / Referral	Reached wrong person

Scenario Templates

Template	Name	Trigger
T15	Voicemail Follow-Up	Within 30 min of voicemail
T16	Meeting Confirmation	Immediately after booking
T17	Post-Call Recap	Within 2 hours of call

Token System

Each template uses `{{token}}` placeholders that the agent fills from lead/research data:

Token	Source	Example
<code>{{first_name}}</code>	<code>contact.first_name</code>	"Sarah"
<code>{{company_name}}</code>	<code>company.name</code>	"Acme Corp"
<code>{{event_name}}</code>	Research <code>events_attended</code>	"Dreamforce 2025"

{{event_date_phrase}}	Research events_attended	"mid-September"
{{days_out}}	Computed from event date	"43"
{{news_headline_short}}	Research recent_news	"your Series B"
{{news_event_implication}}	LLM-generated from news	"growth-stage companies typically double their event calendar"
{{company_vertical_or_motion}}	Enrichment industry	"enterprise SaaS"
{{sender_first_name}}	settings / SenderAccount	"Snehdeep"
{{sender_calendar_link}}	settings	Calendly URL

Sequence Suppression Rules

- After T4 (break-up email), no new sequence starts within **90 days**
- **Maximum 4 emails** in any active sequence
- Reply received at any step → all remaining sequence steps route to reply handlers
- "Not now" reply → log revisit window, resume outreach at that time

Email Output Format

```
{
  "subject": "Cvent build plan for Dreamforce 2025?",
  "body_html": "<html>...</html>",
  "body_text": "Hi Sarah,\n\nDreamforce in mid-September...",
  "template_used": "T8",
  "personalization_hooks": [
    "event: Dreamforce 2025 (31-120 days)",
    "role: sponsor",
    "template: T8"
  ]
}
```

10. Phase 6 — Campaign Execution & Email Delivery

Campaign Structure

A campaign contains:

- **Sequence steps** — ordered array of { step, delay_days, condition, subject_hint } objects
- **Sender account** — which FROM email and display name to use
- **Status** — draft / active / paused / completed

Standard 4-step sequence:

```
Step 0: Day 0 (immediate)
Step 1: Day 4 (delay_days: 4, condition: "no_reply")
Step 2: Day 9 (delay_days: 5, condition: "no_reply")
Step 3: Day 20 (delay_days: 11, condition: "no_reply")
```

Campaign Lead Processing (`process_campaign_lead`)

For each lead in the campaign:

1. **Check reply condition** — if step has `condition: "no_reply"` and a reply exists → skip, mark completed
2. **Fetch full context** — Lead, Contact, Company, AllInsights, ResearchData, EnrichmentData
3. **Resolve sender** — reads `SenderAccount` linked to campaign; falls back to `settings.sender_first_name`
4. **Handle Test Mode** — if tenant has test mode enabled with test email addresses, round-robins delivery to test emails (stored in `personalization_data.test_email_override`); production contacts untouched
5. **Run PersonalizationAgent** — generates subject + body for this lead + step
6. **Create Message record** — `status= "draft"` , `ai_generated=true`
7. **Schedule next FollowUp** — creates `follow_ups` record with `scheduled_at = now() + delay_days`
8. **Dispatch `send_email` task** — auto-queues draft for immediate delivery

Email Delivery (`send_email task`)

1. Resolves recipient email from message record → contact → fallback
2. Resolves sender from `SenderAccount` or environment defaults
3. Sends via SendGrid API (HTML + plain text multipart)
4. Records `X-Message-Id` for tracking
5. Creates `EmailEvent` record (`event_type="sent"`)
6. Sets `message.status = "sent"` , `message.sent_at = now()`

Fallback behavior (no SendGrid key set): logs warning, assigns mock ID, marks sent — safe for local development.

Follow-Up Scheduling (`process_follow_ups`)

A periodic Celery beat task runs on the configured interval and:

1. Queries `follow_ups` table for `status="scheduled"` and `scheduled_at <= now()`
2. For each due follow-up, dispatches `process_campaign_lead.delay(campaign_lead_id)`
3. Marks follow-up as `processed`

11. Orchestration & Pipeline Control

Celery Configuration

Item	Value
Broker	Redis
Task queues	default, enrichment, email

Enrichment queue	run_enrichment_pipeline
Max retries	3 (enrichment), 3 (email)
Retry delay	60s (enrichment), 60s (email), 30s (campaign lead)
Beat scheduler	process_follow_ups periodic task

Enrichment Task Flow

```
@celery_app.task(bind=True, max_retries=3, queue="enrichment")
def run_enrichment_pipeline(self, lead_id, tenant_id, job_ids):
    # job_ids = {"web_research": "uuid", "company": "uuid", "scoring": "uuid"}

    # Set lead enriching
    lead.enrichment_status = "enriching"

    # Step 1 – Research
    job_web_research.status = "running"
    research_output = await ResearchAgent().run(company_name, domain, contact_name)
    persist(ResearchData, AIInsight[research_summary])
    job_web_research.status = "completed"

    # Step 2 – Enrichment
    job_company.status = "running"
    enrichment_output = await EnrichmentAgent().run(raw_data, research_output)
    persist(EnrichmentData, AIInsight[company_enrichment])
    back_fill(company.industry, company.description, contact.department,
contact.title)
    job_company.status = "completed"

    # Step 3 – Scoring
    job_scoring.status = "running"
    score_output = await ScoringAgent().run(lead_data, enrichment_output,
research_output)
    persist(LeadScore, AIInsight[lead_score])
    lead.status = "scored"
    job_scoring.status = "completed"

    # Finalize
    lead.enrichment_status = "enriched"
    lead.enriched_at = now()
```

Error Handling

- Each step catches exceptions independently — a failed research step does not block enrichment or scoring
 - Failed jobs record `job.error = str(exception)` for debugging
 - Pipeline-level failures trigger Celery retry up to 3 times with 60s delay
 - All errors visible in the UI via the Enrichment tab on the Lead detail page
-

12. AI Infrastructure & Model Configuration

LLM Layer (BaseAgent)

All agents extend `BaseAgent` which provides three pre-configured LLM clients:

Method	Model	When Used
<code>get_llm()</code>	claude-sonnet-4-6 (primary)	Research, enrichment, scoring, personalization
<code>get_fast_llm()</code>	claude-sonnet-4-6 (fast model)	Lightweight classification tasks
<code>get_fallback_llm()</code>	gpt-4o / gpt-4o-mini (OpenAI)	Fallback if Claude unavailable

Retry Logic

`BaseAgent.invoke_with_retry()` wraps all LLM calls with:

- **3 attempts max**
- **Exponential backoff:** starts at 2s, max 30s

Temperature Settings by Task

Agent	Temperature	Rationale
ResearchAgent	0.3	Some creativity for synthesis, but grounded
EnrichmentAgent	0.2	Precision-focused, minimize hallucination
ScoringAgent	0.2	Deterministic scoring, reproducible
PersonalizationAgent	~0.7	Natural-sounding email copy

Output Parsing

All agents use LangChain's `JsonOutputParser` to enforce structured output. If parsing fails, the raw response is logged and the job is marked failed.

Cost Tracking

`EnrichmentJob` records `tokens_used` and `cost_usd` per job — enabling per-lead and aggregate cost analysis. (Currently populated by agent responses where token counts are available.)

13. Business Logic & ICP Assumptions

Ideal Customer Profile (ICP)

LaunchHouse Events targets organizations that:

1. **Use Cvent** — or are actively evaluating event tech that includes Cvent
2. **Run B2B events** — corporate events, conferences, trade shows, summits, product launches
3. **Have dedicated event staff** — 1+ event manager/coordinator who builds in Cvent

4. **Experience capacity constraints** — their in-house team can't handle all builds, especially during peak periods or for complex configurations
5. **Are mid-market to enterprise** — typically 50–5,000 employees; too large for DIY, too small / budget-conscious for full-retainer agencies

Vertical Priority (incorporated into scoring)

High-fit verticals (companies in these sectors run the most Cvent-powered events):

- Technology / SaaS (conferences, summits, user days)
- Financial Services (investor events, roadshows, compliance training)
- Pharmaceutical / Healthcare (HCP educational events, congresses)
- Associations & Nonprofits (annual conferences, chapter events)
- Professional Services (client events, thought leadership forums)

Contact Persona Priority

Persona	Titles	Why They Matter
Events Lead (primary)	Head of Events, Director of Events, VP Events	Owns Cvent, selects vendors, direct budget
Marketing Ops	Marketing Operations Manager, Director of Demand Gen	Often manages Cvent license/integration
IT / Procurement	IT Director, Technology Manager	Cvent platform owner in large orgs
C-Suite	CMO, COO	Final sign-off, especially SMB

Cvent-Specific Signal Scoring Logic

The "Event Usage (Cvent)" signal (weight: 10%) scores highest when:

- Company is listed as Cvent customer/case study
- Contact is active in the Cvent Community forums
- Job postings mention "Cvent" as required skill
- Public event registration pages are Cvent-powered (cvent.com/events/...)
- Company attended Cvent Connect (Cvent's annual conference)

"Upcoming Event" Priority Logic

The most commercially valuable trigger is an upcoming event — especially in the **31–120 day window** (T8 template). This is because:

- Event builds in Cvent take 2–8 weeks for anything non-trivial
- The 31–120 day window is when the decision to outsource gets made
- Inside 30 days → rush work only (scoped differently, T7)
- Past 120 days → pipeline nurture, not urgency-based close (T5)

14. Scoring Reference Card

Quick Lookup — Signal Weights

SCORING SIGNAL WEIGHTS	
Company Size Fit	15%
Title / Seniority Fit	15%
Industry Fit	12%
Recent Activity Signals	12%
Technology Fit	10%
Event Usage (Cvent)	10%
Funding / Growth	8%
Engagement History	8%
Timing Signals	5%
Geographic Fit	5%
TOTAL	100%

Tier Color Reference

Tier	Score	Action
 Hot	≥ 75	Prioritize — direct sales motion
 Warm	50–74	Personalized nurture sequence
 Cold	< 50	Low-touch future nurture or skip

15. Template Reference Card

Template Selection Quick Reference

Situation	→ Template
Positive reply	→ T10
"Send more info" reply	→ T11
"Not now / later" reply	→ T12
"Already have vendor" reply	→ T13
Wrong contact	→ T14
Event verified, WITHIN 30 days	→ T7 (Rush)
Event verified, 31–120 days	→ T8 (Build – best)
Event verified, 120+ days/past	→ T5 (Event trigger)
News ≤45 days + events	→ T9 (News trigger)
Activity inferable, no event/news	→ T6 (Fit-based)
Step 1 (no other context)	→ T1 (Initial)
Step 2 (no reply yet)	→ T2 (Value add)
Step 3 (no reply yet)	→ T3 (Bump)
Step 4 (no reply yet)	→ T4 (Break-up)
Post-voicemail (<30 min)	→ T15

Meeting booked (immediate)	→ T16
Post-call (<2 hours)	→ T17

Core Sequence Timing

Day 0: T1 – Initial Outreach
Day 4: T2 – Value Add Follow-Up
Day 9: T3 – Bump ("worth a reply?")
Day 20: T4 – Break-Up (→ moves to 90-day nurture)

Suppression: 90 days after T4 before new sequence. Max 4 emails per sequence.

16. Current Limitations

Research Quality

- **Live web search is active.** Tavily is the primary provider (4–5 queries per lead), with Firecrawl search as fallback and Firecrawl homepage scrape always running when a domain is available.
- **Knowledge recency depends on Tavily index freshness.** Very recently published pages (hours old) may not yet be indexed.
- **No direct Cvent event calendar API.** Upcoming events are found via web search results and homepage scrapes, not by querying Cvent's platform directly.

Scoring Accuracy

- Heavy dependence on title/industry classification from raw data or enrichment output
- Engagement History signal (8%) is mostly 0 for new leads with no prior interaction
- Geographic Fit uses coarse inference (company location from `contact.location` or `company.location`)

Personalization

- Token filling is LLM-generated — hallucinated event names or news are possible (especially with stubs returning nothing)
- `{{days_out}}` calculation requires structured `events_attended` data with accurate future dates — quality depends on research accuracy

Email Infrastructure

- `message.message_id` field is reused (dual-purpose: stores recipient email during draft creation, then overwritten with SendGrid message ID post-send) — a technical debt item
 - Reply classification (`intent` field on `Reply` model) is currently set externally — no automated reply parsing is implemented in the current codebase
-

17. Future Enhancements

High Priority

1. **Live web search integration**  *Completed April 2026*
 - Tavily (primary) and Firecrawl (fallback + homepage scrape) are wired and live
 - Remaining: real-time Cvent.com event calendar scraping via direct API
 - Remaining: LinkedIn Sales Navigator API for contact verification

2. Automated reply classification

- NLP pipeline to parse inbound replies and set `intent` automatically
- Categories: positive, not-now (with timeframe extraction), wrong-contact, unsubscribe, question, bounce
- Would enable fully automated campaign branching without human review

3. Custom scoring profiles

- Allow per-tenant or per-campaign signal weight customization
- Enable vertical-specific profiles (e.g., different weights for Pharma vs. SaaS)

4. Cvent Connect attendee data

- Direct integration with Cvent's attendee export for event participant verification
- Would elevate Event Usage signal accuracy significantly

Medium Priority

5. A/B template testing

- Track open/reply rates per template variant
- Automated winning variant selection
- Long-form vs. short-form performance tracking

6. ICP profile refinement loop

- Feed back won/converted leads to refine scoring weights
- Automatic weight adjustment based on conversion data

7. Enrichment refresh scheduling

- Auto-re-enrich leads after 90 days
- Trigger re-enrichment on company funding news, leadership change, or job posting activity

8. Multi-sender rotation

- Assign leads across multiple `SenderAccount` records for volume scaling
- Warmup tracking per sender domain

Lower Priority

9. LinkedIn outreach channel

- Extend `Message` model to support LinkedIn DMs alongside email
- Coordinated multi-channel sequences

10. Webhook-based real-time triggers

- Cvent webhook for registration spike detection mid-sequence
- HubSpot/Salesforce synced enrichment for CRM-first workflows

11. Custom field scoring rules

- Allow admins to define scoring logic for `custom_fields` data
 - E.g., "if custom_field.cvent_tier = 'enterprise', add 10 points to Event Usage signal"
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Document generated from source code analysis. Last updated: April 25, 2026.

Backend: `backend/app/agents/` , `backend/app/tasks/` , `backend/app/models/lead.py` ,
`backend/app/tools/web_search.py`